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Case 31-2025: A 56-Year-Old Man with Left Lower Abdominal Pain and Anemia

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PRESENTATION OF CASE

Dr. Brooke A. Rice (Medicine): A 56-year-old man was admitted to this hospital because of left lower abdominal pain and anemia.

The patient had been in his usual state of health until 1 year before the current presentation, when diarrhea developed. The stools were watery, foamy, and non-bloody. The diarrhea was worst after he had eaten, although it did not fully resolve with fasting. It occasionally woke him up from sleep and was associated with bloating but no pain, nausea, or vomiting.

Eleven months before the current presentation, the patient was evaluated at another hospital. Computed tomography (CT) of the abdomen and pelvis reportedly revealed thickening of proximal jejunal loops. A possible fistula between a proximal jejunal loop and the distal transverse colon was noted. Esophagogastroduodenoscopy (EGD) and colonoscopy reportedly showed scalloped mucosa in the second portion of the duodenum and normal mucosa in the colon. A biopsy specimen from the duodenum reportedly showed villous blunting and intraepithelial lymphocytosis; a biopsy specimen from the colon reportedly showed glandular dysplasia. Testing for IgG and IgA to tissue transglutaminase was negative. The IgA level was 9 mg per deciliter (reference range, 47 to 310). A diagnosis of celiac disease was considered, and treatment with a gluten-free diet was started. Bloating abated, but diarrhea persisted.

Ten months before the current presentation, the patient was evaluated in the outpatient gastroenterology clinic affiliated with the other hospital because of persistent diarrhea. The laboratory evaluation revealed iron-deficiency anemia and a fecal calprotectin level of 493 μ g per gram (reference value, <50). Stool studies were positive for giardia antigen; a course of tinidazole was prescribed.

During the next 8 months, several stool studies were positive for giardia antigen, and courses of antimicrobial agents — including metronidazole, nitazoxanide, and metronidazole plus albendazole — were administered. The diarrhea abated

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briefly with each course and then recurred. A screening test for human immunodeficiency virus types 1 and 2 was negative.

Two months before the current presentation, the patient was admitted to this hospital because of persistent diarrhea and weight loss of 23 kg in the past year. The level of IgG to giardia in stool was 4.6 U per milliliter (reference value, <20.6); treatment with nitazoxanide was started and was continued for 7 days. The blood levels of C-reactive protein, cobalamin, folate, vitamin E, and vitamin A were normal. The copper level was 65 μ g per deciliter (reference range, 73 to 129), the zinc level 49 μ g per deciliter (reference range, 60 to 106), the vitamin K level 0.4 ng per milliliter (reference range, 0.1 to 2.2), and the 25-hydroxyvitamin D level 11 ng per milliliter (reference range, 20 to 80).

Dr. Anuradha S. Shenoy-Bhangle: CT of the abdomen and pelvis (Fig. 1A and 1B) revealed diffuse mural thickening and mucosal hyperenhancement of the colon. Several jejunal loops were tethered to the distal transverse colon. Numerous prominent mesenteric lymph nodes were noted; the spleen was normal in size. Magnetic resonance enterography (Fig. 1C and 1D) showed an enterocolonic fistula between the proximal jejunum and the distal transverse colon. Short-segment wall thickening and a focal stricture of the involved jejunal loop were causing mild upstream jejunal dilatation. No associated high-grade bowel obstruction was present.

Dr. Rice: EGD was performed along with push enteroscopy and colonoscopy. Diffuse abnormal nodular mucosa was present in the duodenum and jejunum. The fistula in the proximal jejunum was surrounded by ulceration and friable tissue, with no clear mass. Colonoscopy showed a 1-mm fistula in the transverse colon with surrounding ulceration and a normal-appearing terminal ileum.

Dr. Dingani Nkosi: Biopsy specimens from the duodenum and jejunum (Fig. 2) showed variably active neutrophilic enteritis. A lymphoid inflammatory infiltrate was seen in the lamina propria, as was a decrease in plasma cells. Rare organisms that were consistent with giardia were also identified in the duodenal specimen. Ulceration with mucosal granulation tissue and fibrinopurulent exudate was identified in the jejunal specimen, without evidence of cancer. Immunohistochemical staining for cytomegalovirus and

adenovirus was negative in the small-bowel specimens, and staining for CD117 highlighted mast cells with normal morphologic features and a normal distribution.

Biopsy specimens from the area of the colonic fistula showed fragments of organizing granulation tissue and fibrinopurulent exudate, findings consistent with ulceration in the presence of a fistula tract. Special staining for infectious microorganisms — including periodic acid–Schiff staining, Grocott's methenamine silver staining, Steiner staining, and acid-fast staining — was negative. The finding of sparse plasma cells in the lamina propria of enterocolonic mucosa was consistent with immunoglobulin deficiency.

Dr. Rice: The patient had negative tests for IgG, IgA, and IgM to enterocytes; IgG and IgA to gliadin; and IgG and IgA to tissue transglutaminase. The test for antibodies to tissue transglutaminase was performed while he was following a gluten-free diet. Testing for the presence of the HLA-DQ2 and HLA-DQ8 haplotypes was also negative, as was a nucleic acid amplification test of blood for *Tropheryma whipplei* DNA.

Additional testing for immunodeficiency was performed. The blood levels of IgM and IgA were undetectable. The IgG level was 1190 mg per deciliter (reference range, 614 to 1295), with a normal level of IgG1 but with low levels of IgG2, IgG3, and IgG4. Serum electrophoresis with immunofixation showed an IgG lambda M component in the gamma region (0.81 g per deciliter). A diagnosis of monoclonal gammopathy of undetermined significance was suspected. Flow cytometry performed on peripheral blood revealed an immunophenotypically abnormal natural killer (NK)-cell population with features suggestive of clonality. A peripheral-blood smear showed a subpopulation of monotonous-appearing, slightly enlarged lymphoid cells with round-to-indented nuclei, condensed chromatin, and abundant eccentric pale granular cytoplasm. Sequencing of a blood sample with the use of a rapid heme panel was normal. The expanded NK-cell population was thought to be caused by either autoimmunity or infection; plans were made to perform repeat flow cytometry in 6 months to reassess the NK-cell population.

A diagnosis of Crohn's disease was suspected. The patient was hospitalized for approximately 1 month and received 17 days of treatment with

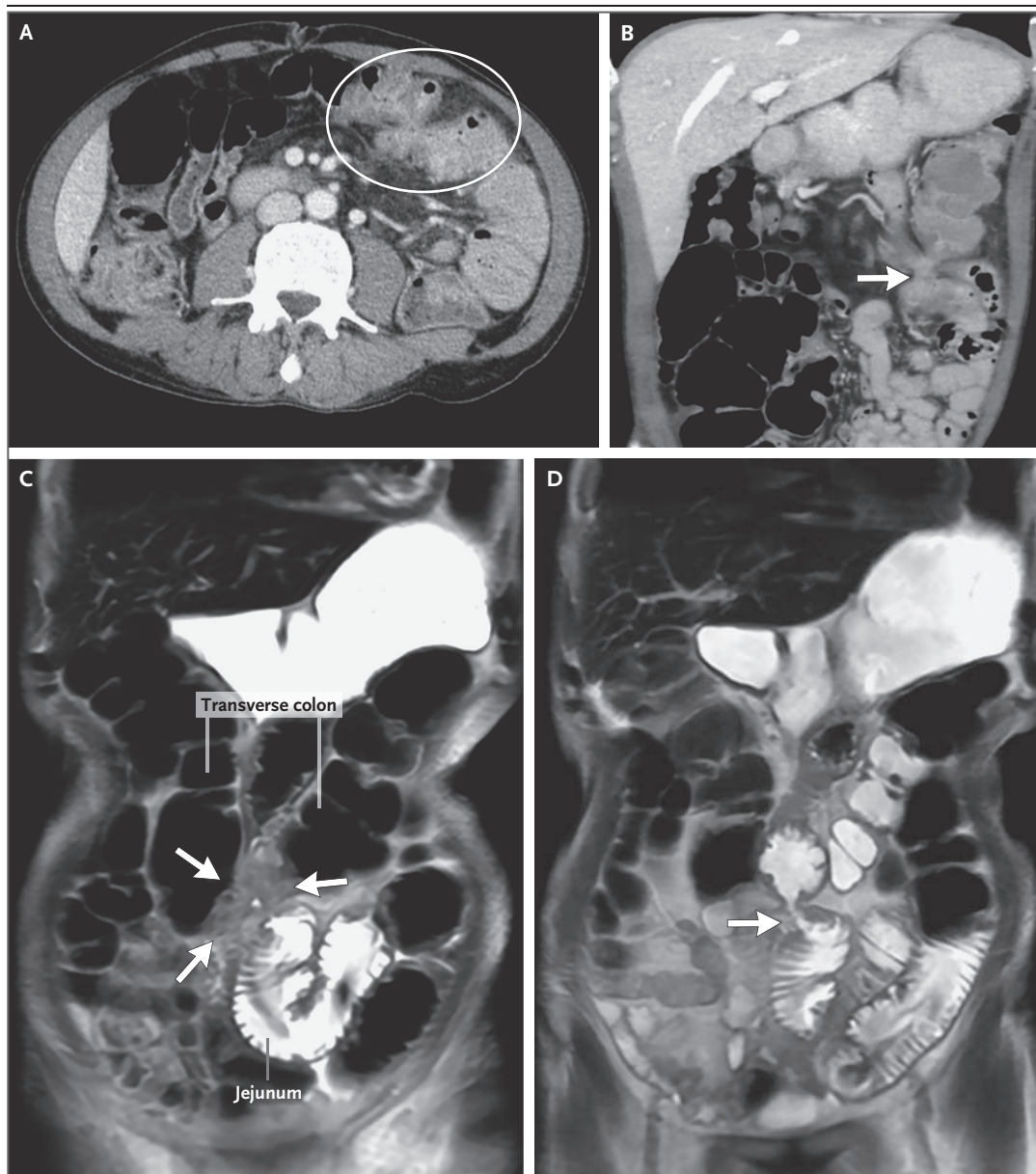


Figure 1. Abdominal Imaging Studies.

CT of the abdomen was performed with the administration of contrast material in the portal venous phase. An axial image (Panel A) shows concentric wall thickening of a long segment of jejunal loop in the left upper quadrant (circled). A coronal image (Panel B) shows a possible fistula between the jejunum and the transverse colon (arrow). Magnetic resonance enterography was performed subsequently. Coronal half-Fourier single-shot turbo spin-echo images (Panels C and D) confirm the presence of a fistula (Panel C, arrows) and show a focal stricture in the affected jejunal loop (Panel D, arrow).

total parenteral nutrition, which was tapered as his oral intake of food increased. Diarrhea resolved during the admission. Treatment with infliximab was started, with the first dose administered 2 days before discharge. In the presence of hypogammaglobulinemia, two doses of

intravenous immune globulin were administered before immunosuppression with infliximab was initiated.

After discharge, diarrhea did not recur, and the patient was eating normally. He returned to his work in construction.

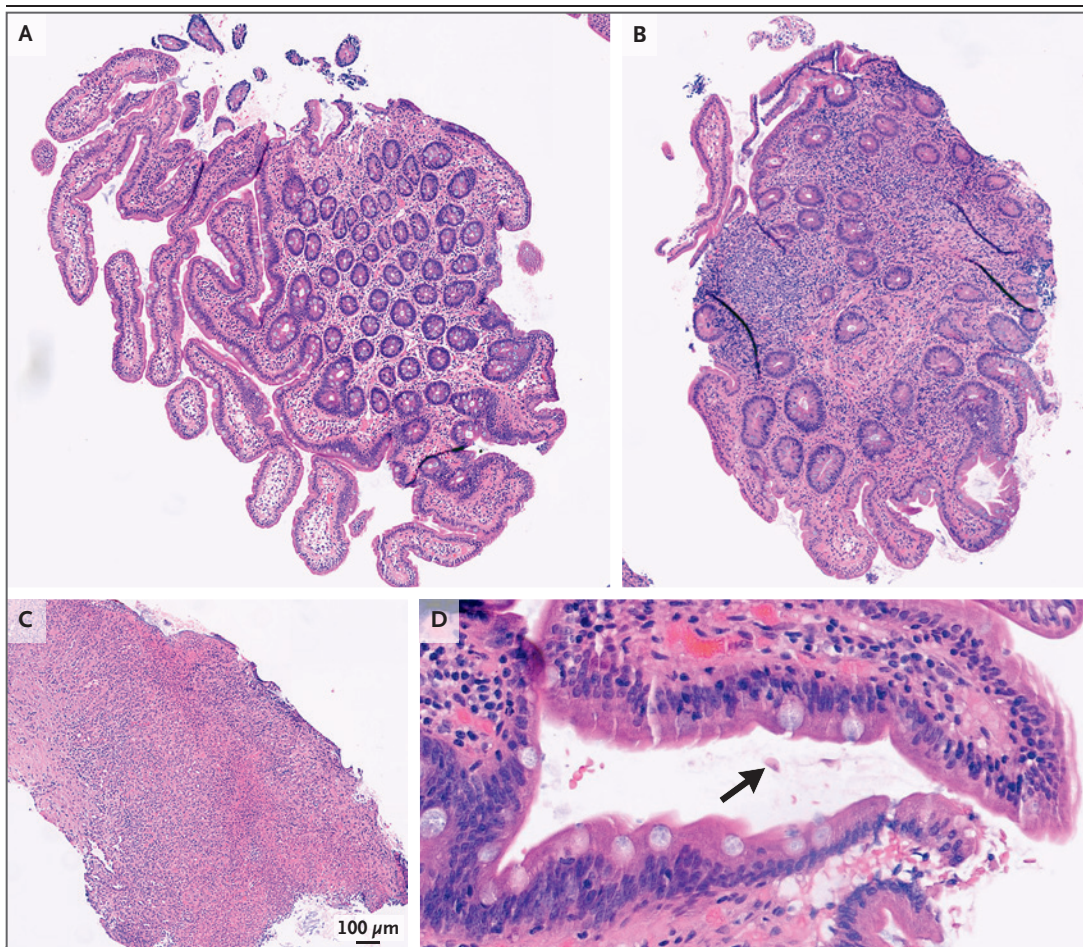


Figure 2. Biopsy Specimens of the Duodenum and Jejunum.

On hematoxylin and eosin staining, the duodenal-biopsy specimen shows normal villi and a lymphoid inflammatory infiltrate in the lamina propria (Panels A and B). The jejunal-biopsy specimen shows ulceration with mucosal granulation (Panel C). The duodenal-biopsy specimen also shows rare organisms that are consistent with giardia (Panel D, arrow).

Ten days before the current presentation, dull pain in the left lower abdomen developed. During the next 10 days, stools were intermittently dark, without hematochezia or melena. Two days before the current presentation, outpatient laboratory testing revealed a hemoglobin level of 6.8 g per deciliter (reference range, 13.2 to 17.1); the level had been 8.0 g per deciliter on discharge from this hospital 1 month earlier. The patient's primary care physician recommended that he present to the emergency department of this hospital for evaluation.

In the emergency department, the patient reported occasional lightheadedness with position changes and ongoing mild left lower abdominal

pain. Additional medical history included nephrolithiasis and dyslipidemia; surgical history included appendectomy and cholecystectomy. In addition to infliximab, medications included cholecalciferol, cyanocobalamin, ferrous sulfate, and a multivitamin. There were no known adverse reactions to medications. The patient was born in Brazil and had lived in the United States for 27 years. He had last visited Brazil 7 years before the current presentation. He lived with his wife and two teenage children in coastal Massachusetts. Their home had a city water supply. He was a lifelong nonsmoker and did not drink alcohol or use drugs. His sister had died at 44 years of age from an unknown cause.

On examination, the temporal temperature was 36.6°C, the blood pressure 113/65 mm Hg, the pulse 84 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 24.0; the weight had decreased by 4 kg in the month since he was discharged from this hospital. The left lower abdomen was tender on palpation. The ankles had trace edema. No external hemorrhoids were noted, and the rectal tone was normal. No stool or gross blood was seen on a digital rectal examination.

The blood levels of electrolytes, glucose, alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, bilirubin, and lactic acid were normal, as were the results of coagulation and kidney-function tests. The hemoglobin level was 7.5 g per deciliter (reference range, 13.5 to 17.5); the complete blood count and differential count were otherwise normal.

Dr. Shenoy-Bhangle: CT angiography of the abdomen and pelvis (Fig. 3) showed no evidence of active bleeding but revealed marked concentric wall thickening and aneurysmal dilatation of two discontinuous segments of proximal jejunum located in the left upper abdomen. The segments had mural hypoenhancement that was suggestive of bowel-wall ischemia. Adjacent proximal jejunal loops had mucosal hyperenhancement.

Dr. Rice: The patient was admitted to the hospital. A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Bruce E. Sands: This 56-year-old man who was originally from Brazil had received provisional diagnoses of celiac disease and IgA deficiency after presenting with nonbloody diarrhea. Diarrhea persisted despite adherence to a gluten-free diet, and in the ensuing months, additional evaluation revealed chronic giardiasis, IgM deficiency, and laboratory evidence of malnutrition and also confirmed the presence of an enterocolonic fistula. He was treated with tumor necrosis factor α (TNF- α) inhibitor therapy for suspected Crohn's disease before presenting with worsening anemia and weight loss. The patient's confusing array of unusual findings evolved over

time, leading to changes in the provisional diagnoses that could explain the abnormalities. Therefore, it is worthwhile to consider each of the provisional diagnoses, as well as the key findings that require further explanation to arrive at the correct diagnosis.

CELIAC DISEASE

Celiac disease was diagnosed after scalloped mucosa was seen in the duodenum on EGD, with villous blunting and intraepithelial lymphocytosis seen on histopathological examination. However, serologic testing for antibodies to tissue transglutaminase was negative — specifically testing for IgA to tissue transglutaminase, which is more accurate in the diagnosis of celiac disease than testing for IgG to tissue transglutaminase.¹ Although the concurrent finding of IgA deficiency might have led to false negative testing, several other clues pointed toward a diagnosis other than celiac disease.

The diarrhea did not resolve with adherence to a gluten-free diet. Moreover, the CT findings — the thickened proximal jejunal loops and, in particular, the possible fistula between the proximal jejunum and the transverse colon — were not consistent with a diagnosis of celiac disease. Indeed, the patient later had negative testing for the presence of the HLA-DQ2 and HLA-DQ8 haplotypes, which are associated with celiac disease, a finding that provided convincing evidence against the diagnosis.¹

IGA DEFICIENCY

Selective IgA deficiency is the most common immunodeficiency, affecting approximately 1 in 500 persons in North America.² It is usually an incidental laboratory finding; most patients are asymptomatic.² When IgA deficiency is detected, quantitative immunoglobulin testing is performed to determine whether IgM and IgG deficiencies are also present. In this patient, the chronic giardiasis was suggestive of clinically significant immunodeficiency. Ten months after the initial discovery of IgA deficiency, additional testing revealed IgM deficiency.

SMALL-BOWEL FISTULA

The patient's initial CT imaging, performed at the other hospital, reportedly showed a possible fistula between a proximal jejunal loop and the

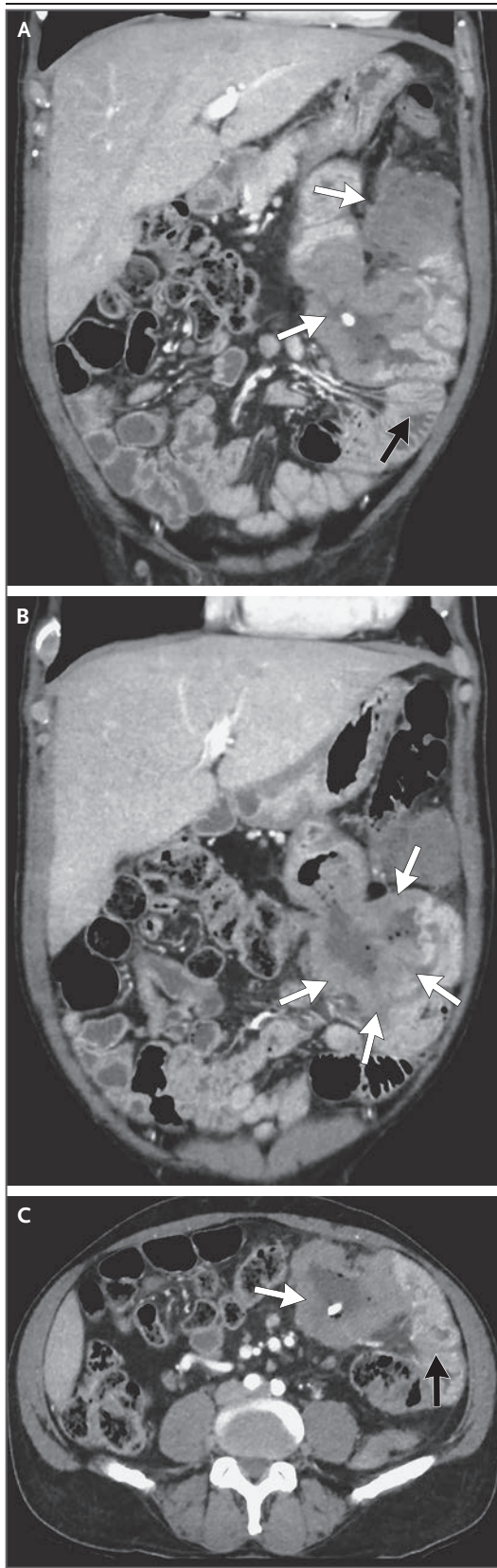


Figure 3. CT Angiograms.

Coronal images (Panels A and B) and an axial image (Panel C) show a hypoenhancing jejunal loop (white arrows) with marked thickening and aneurysmal dilatation, without bowel obstruction. Adjacent hyperenhancing jejunal loops are also shown (Panels A and C, black arrows).

transverse colon. Magnetic resonance enterography is more sensitive than CT for the detection of intestinal fistulas³; 9 months after the initial CT imaging, magnetic resonance enterography performed at this hospital confirmed the presence of an enterocolonic fistula.

Fistulas arising from the small bowel are uncommon and have a limited differential diagnosis.⁴ Crohn's disease is the most common cause of small-bowel fistulas, but jejunal involvement occurs in just 3% of patients with Crohn's disease,⁵ whereas ileal involvement occurs in nearly two thirds of patients. Jejunal involvement resulting in a fistula is even less common in patients with Crohn's disease.⁵ Therefore, other possible causes of an enterocolonic fistula are important to consider.

Foreign-body ingestion, diverticulosis, and ischemia can cause small-bowel fistulas. CT imaging performed during the patient's first admission to this hospital did not show findings suggestive of ischemia or diverticular disease of the small bowel. Foreign bodies, such as an ingested toothpick, are not always radiopaque, but the presence of an isolated foreign body would not typically be associated with bowel-wall thickening and mucosal enhancement, which are findings consistent with extensive jejunal inflammation.

Autoimmune enteropathy is an unlikely diagnosis in this case, given the negative test for antibodies to enterocytes and the absence of marked epithelial destruction on histopathological examination of jejunal-biopsy specimens. In addition, tests were negative for a variety of unusual enteric pathogens that are associated with small-bowel fistulas, including special staining for infectious microorganisms and nucleic acid amplification testing for *T. whipplei*.

Intestinal cancers such as small-bowel adenocarcinoma, lymphoma, and other lymphoproliferative disorders can cause small-bowel fistulas. Endoscopic biopsy specimens obtained from the area where obvious jejunal inflammation was

noted on CT showed a mixed lymphohistiocytic infiltrate with some active neutrophilic enteritis. However, the infiltrate in the lamina propria was not consistent with a monotonous clonal infiltrate, and no epithelial destruction was noted. Therefore, despite the abnormal clonal NK-cell population in peripheral blood, which was thought to indicate monoclonal gammopathy of undetermined significance, no evidence of an intestinal cancer was found during the patient's first admission to this hospital.

The inflammatory infiltrate seen on histopathological evaluation of the jejunal-biopsy specimen was typical of Crohn's disease except in one way: the apparent lack of plasma cells. This finding suggests that a primary immunodeficiency may be underlying this patient's presentation.

PRIMARY IMMUNODEFICIENCY

Common variable immunodeficiency (CVID) is the most common symptomatic immunodeficiency that develops in adulthood, occurring in up to 1 in 25,000 White persons.² However, several aspects of this patient's presentation would be unusual for CVID.⁶ Symptoms of CVID typically develop in the first or third decade of life. Recurrent respiratory infections affect approximately three quarters of patients with CVID and were not reported in this patient. We are not provided with information about antibody responses to T-dependent or T-independent antigens, such as those found in vaccines; poor antibody responses would be expected in a patient with CVID.

Although the patient had IgA and IgM deficiencies, he did not have a deficiency in total IgG, which is considered to be an essential finding in all proposed diagnostic criteria for CVID. It is interesting that serum electrophoresis with immunofixation showed a monoclonal IgG band. Nevertheless, the patient had deficiencies in all IgG subclasses except for IgG1, and monoclonal gammopathy of undetermined significance may explain the normal level of total IgG. These findings suggest disordered B-cell maturation and isotype switching. Regardless of whether the patient is considered to have CVID or to have an idiopathic primary hypogammaglobulinemia that does not fully meet the criteria for CVID, the immunoglobulin deficiencies, the lack of plasma cells on intestinal biopsy, and the recur-

rent giardiasis suggest a condition on the spectrum of CVID.

Gastrointestinal manifestations are reported in at least 20% of patients with CVID. Such manifestations may include enteric infections; non-infectious chronic enteropathy, which may have clinical or pathological features resembling those of celiac disease; or inflammatory bowel disease, which can result in small-bowel or colonic involvement similar to that seen with Crohn's disease or ulcerative colitis.² Very few cases of CVID enteropathy complicated by a fistula are reported in the literature.⁷ This patient's pathological, endoscopic, and imaging findings most closely resembled those of Crohn's disease. Given the severity of his presentation and the enterocolonic fistula, his physicians' choice to begin treatment with the anti-TNF- α antibody infliximab seems reasonable. Infliximab is effective in treating fistulizing Crohn's disease,⁸ and case reports have described the successful use of infliximab (and other biologic agents) in treating inflammatory bowel disease-like enteropathy arising in CVID.⁹

SECOND HOSPITALIZATION

One month after treatment with infliximab, the patient presented with new left lower abdominal pain and worsening anemia. CT angiography was performed to assess for active gastrointestinal bleeding but instead showed marked wall thickening and dilatation of two segments of proximal jejunum with mucosal hypoenhancement, which suggested ischemia. These findings call into question my earlier assumption that the enterocolonic fistula was not caused by ischemia. I will form new hypotheses in which the intestinal loss of clotting factors from inflamed small bowel might confer a predisposition to thrombosis, although no thrombosis was described in the CT report.

It is worthwhile to consider a dictum of Sir Arthur Conan Doyle, expressed through the persona of Sherlock Holmes: "It is a capital mistake to theorize before one has data. Insensibly one begins to twist facts to suit theories, instead of theories to suit facts."¹⁰ The key facts leading to my diagnosis in this case are found on review of the CT images, which show new and profound thickening of the jejunum, effacing the normal bowel architecture, with marked dilatation. This discovery is a potent reminder to personally re-

view the primary sources of data, which in this case are the CT images. The findings on imaging strongly suggest yet another gastrointestinal manifestation of CVID, namely small-bowel lymphoma.

SMALL-BOWEL LYMPHOMA

In up to 10% of patients with CVID, lymphoid cancer develops, most often non-Hodgkin's B-cell lymphoma.¹¹ Given the chronic immune stimulation associated with recurrent enteric infection, as well as the ongoing enteric inflammation associated with dysregulated mucosal immune responses that occur with CVID, it is not surprising that lymphomas develop at extranodal sites. It is notable that epidemiologic studies have linked treatment with anti-TNF- α antibodies to an increased risk of non-Hodgkin's lymphoma,¹² regardless of whether the disease is related to Epstein-Barr virus infection. The temporal relationship between the patient's current presentation and the initiation of treatment with infliximab suggests the possibility that this treatment was a contributing factor. It is also possible that this treatment unmasked a smoldering intestinal lymphoma that had been present even at the time of the initial presentation. The results of previous histopathological evaluations did not, however, suggest that this was the case.

This patient most likely had a non-Hodgkin's B-cell lymphoma of the jejunum complicating Crohn's-like enteropathy resulting from idiopathic primary hypogammaglobulinemia. The diagnosis should be confirmed by means of push enteroscopy for biopsy and histopathological evaluation of the jejunum.

DR. BRUCE E. SANDS' DIAGNOSIS

Non-Hodgkin's B-cell lymphoma of the jejunum complicating Crohn's-like enteropathy resulting from idiopathic primary hypogammaglobulinemia.

PATHOLOGICAL DISCUSSION

Dr. Nkosi: Jejunal-biopsy specimens were obtained (Fig. 4). Hematoxylin and eosin staining showed diffuse infiltration of large lymphoid cells with vesicular chromatin and associated apoptotic debris. Immunohistochemical staining

showed that the large cells were CD20-positive B cells with variable coexpression of CD10, BCL6, and MUM1; these features are consistent with a diagnosis of diffuse large B-cell lymphoma. The Ki-67 proliferation index was approximately 70 to 80%.

PATHOLOGICAL DIAGNOSIS

Diffuse large B-cell lymphoma.

DISCUSSION OF MANAGEMENT

Dr. P. Connor Johnson: Diffuse large B-cell lymphoma is the most common type of non-Hodgkin's lymphoma, accounting for approximately 25,000 cases in the United States annually.¹³ In this patient's case, we suspect that an underlying immunodeficiency, which is a known risk factor for non-Hodgkin's lymphoma, contributed to the development of the disease.

Diffuse large B-cell lymphoma is an aggressive blood cancer that is fatal if untreated. Treatment with chemotherapy and rituximab, a monoclonal antibody, leads to cure in approximately two thirds of cases.¹⁴ Several treatment regimens are available, and the International Prognostic Index (IPI) score is often used to guide therapy selection.¹⁵ IPI scores range from 0 to 5, with higher scores indicating higher-risk disease with a poorer prognosis. In anticipation of treatment with chemoimmunotherapy, the infectious disease consultant evaluated the patient regarding the persistent positive tests for giardia antigen.

Dr. Jacob E. Lemieux: We could not find literature reporting the duration of positivity for giardia antigen testing, but we doubted that the antigen positivity would have persisted for so long if the giardia had been successfully eradicated. A quantitative giardia polymerase-chain-reaction assay was not available; we performed a review of the histopathological slides as a proxy, and a substantial burden of giardia trophozoites was detected in the lumen of the jejunum. We thus considered the patient to have chronic active giardiasis that was refractory to treatment attempts with multiple agents.

In patients with chronic giardiasis, mucosal immune responses often fail. When this occurs, treatment may involve prolonged courses or com-

ination therapies, some of which had already been prescribed in this patient. We asked the patient whether nonadherence to treatment could be a contributing factor. The patient expressed

that he had concerns about taking too many antibiotics and acknowledged that he had not fully completed all previous courses because of these concerns.

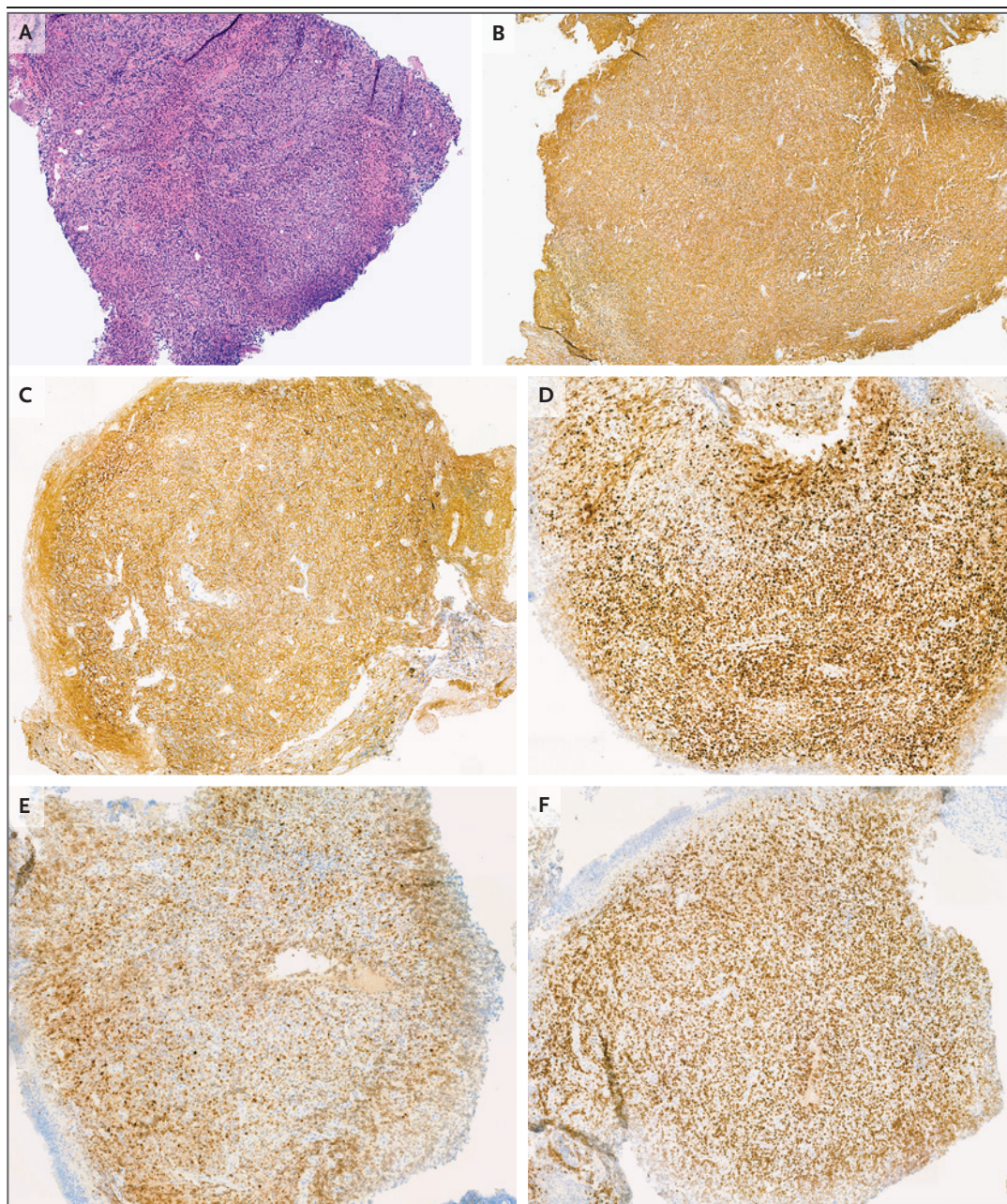


Figure 4. Biopsy Specimens of the Jejunum.

Hematoxylin and eosin staining of representative sections of the jejunum shows diffuse infiltration of large lymphoid cells with vesicular chromatin and associated apoptotic debris (Panel A). Immunohistochemical staining shows that the large cells are CD20-positive B cells (Panel B) with variable coexpression of CD10, BCL6, and MUM1 (Panels C, D, and E, respectively). The Ki-67 proliferation index is approximately 70 to 80% (Panel F).

In selecting the treatment regimen, we considered salvage regimens for refractory giardiasis, such as quinacrine combined with metronidazole or tinidazole.¹⁶ Quinacrine was not available. As an alternative, we prescribed a prolonged course of combination therapy — a 21-day course of albendazole combined with metronidazole — with a plan to monitor for symptoms and, if symptoms were to recur, test for antigen positivity. The patient completed a 7-day course of metronidazole, and at follow-up, he reported resolution of the giardiasis symptoms.

Dr. Johnson: The patient received six cycles of chemoimmunotherapy with rituximab, doxorubicin, cyclophosphamide, vincristine, and prednisone, which was selected on the basis of his low IPI score (<2). He had minimal side effects and was in complete remission at the end of treatment. The patient is now undergoing sur-

veillance monitoring, and a comprehensive immunologic evaluation is planned for the detection of any underlying immunodeficiency.

FINAL DIAGNOSIS

Diffuse large B-cell lymphoma.

This case was presented at the Medicine Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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